

2026

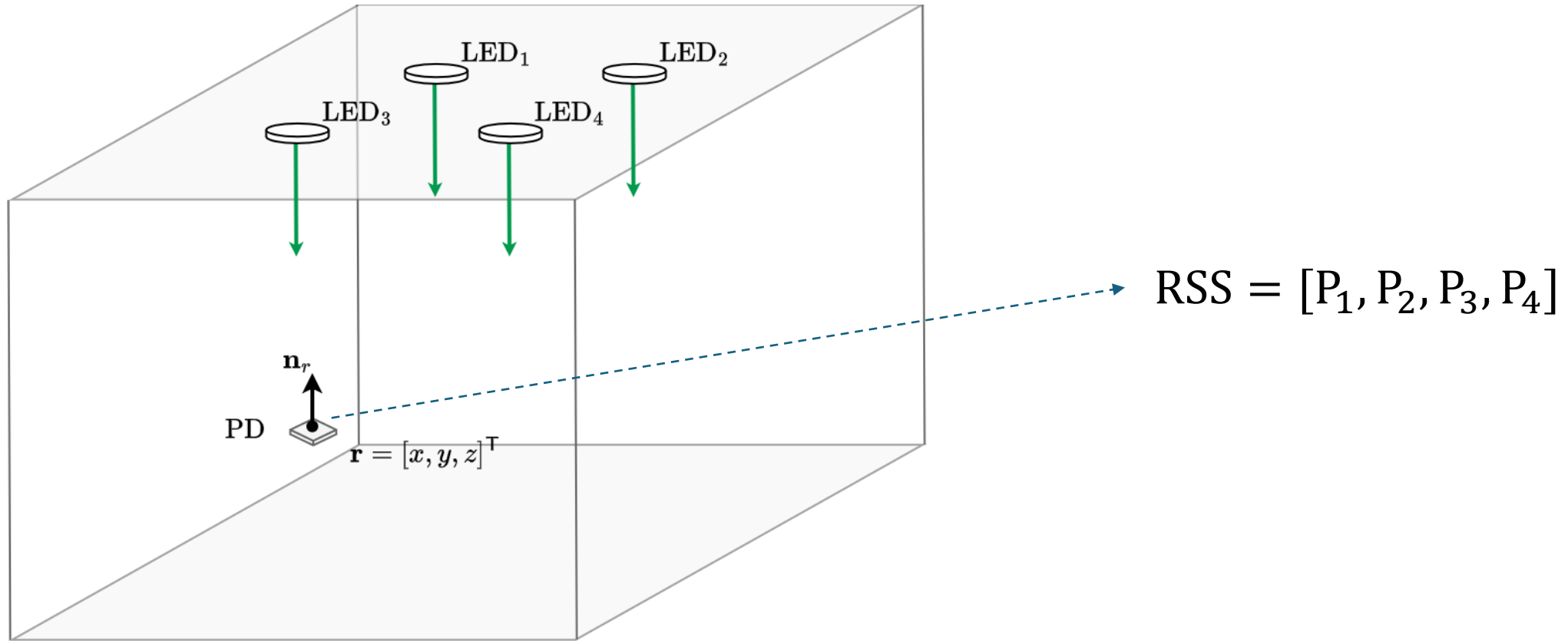
Beam-steered Single-LED VLP for 3D localization

Kevin Jose ACUNA CONDORI

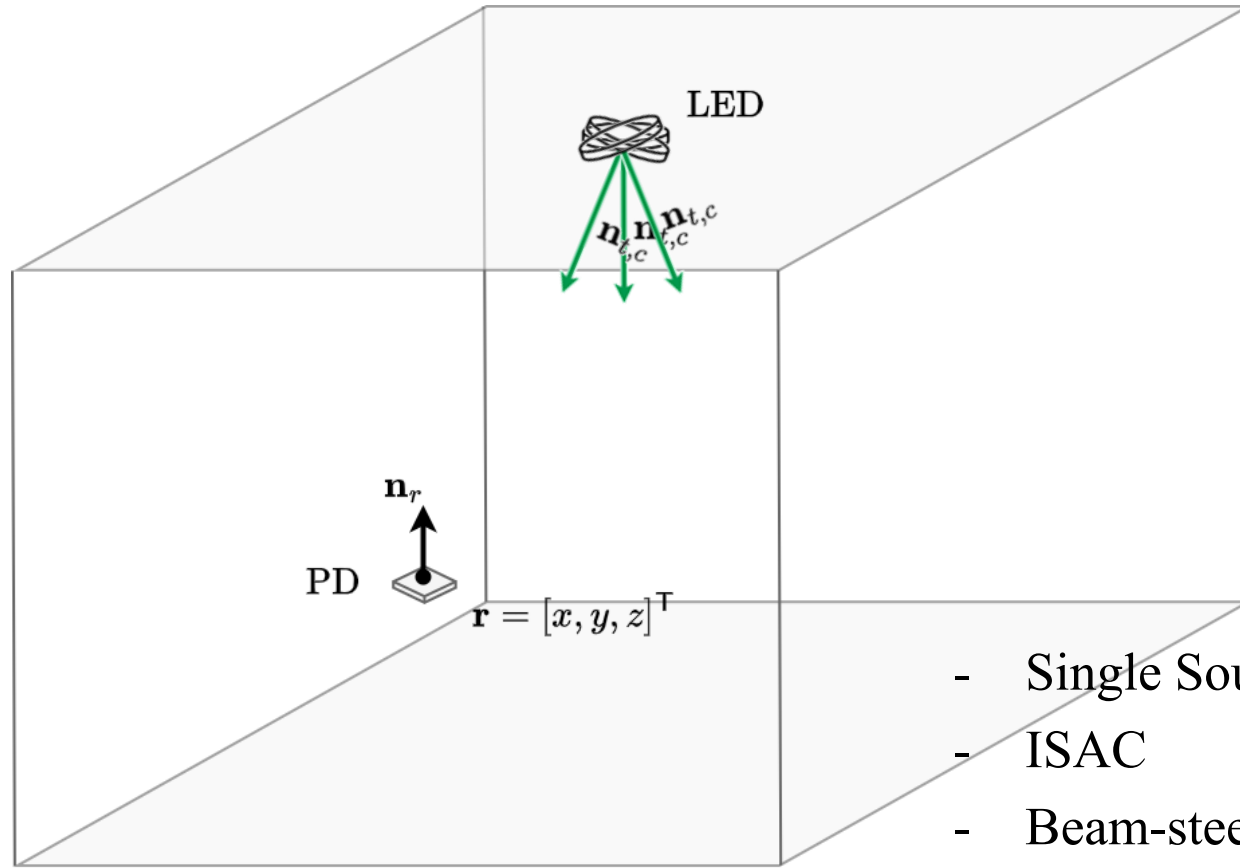
08/09/2026

1) PhD topic

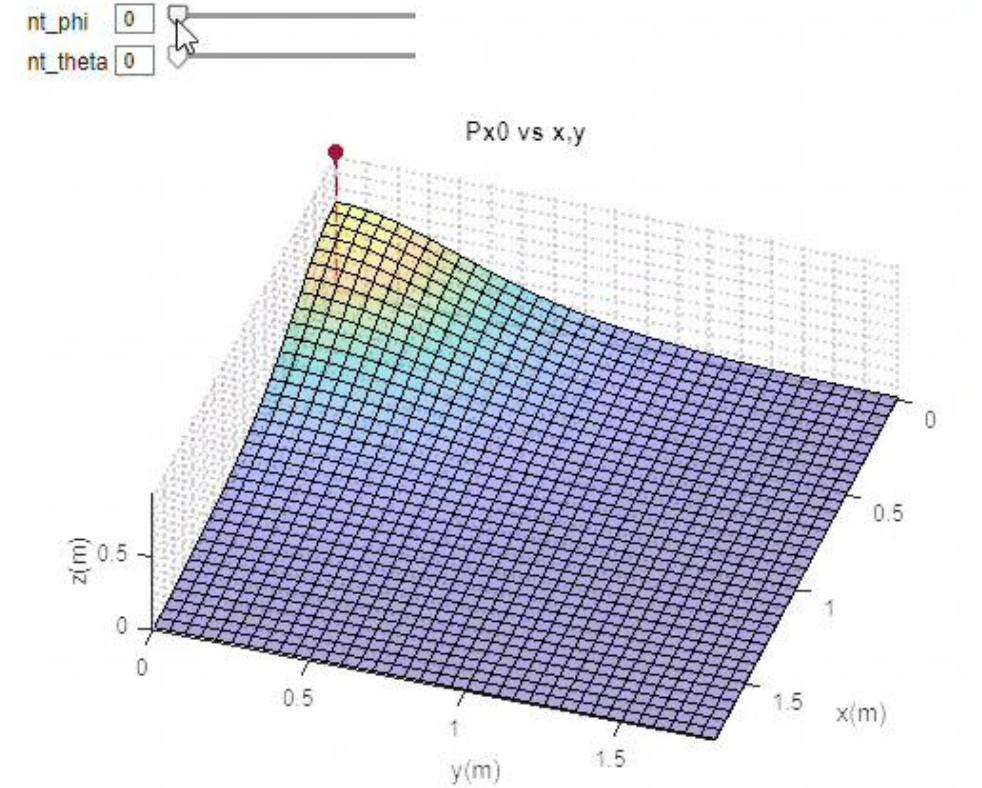
Optical Wireless Positioning



1) PhD topic:

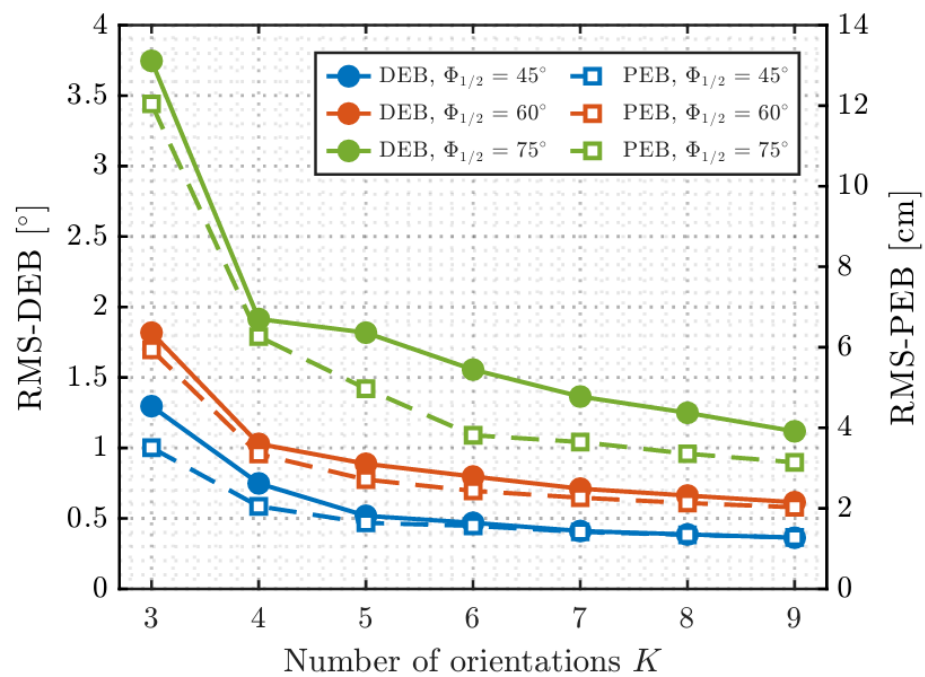


- Single Source
- ISAC
- Beam-steered Source
- Independent of the layout

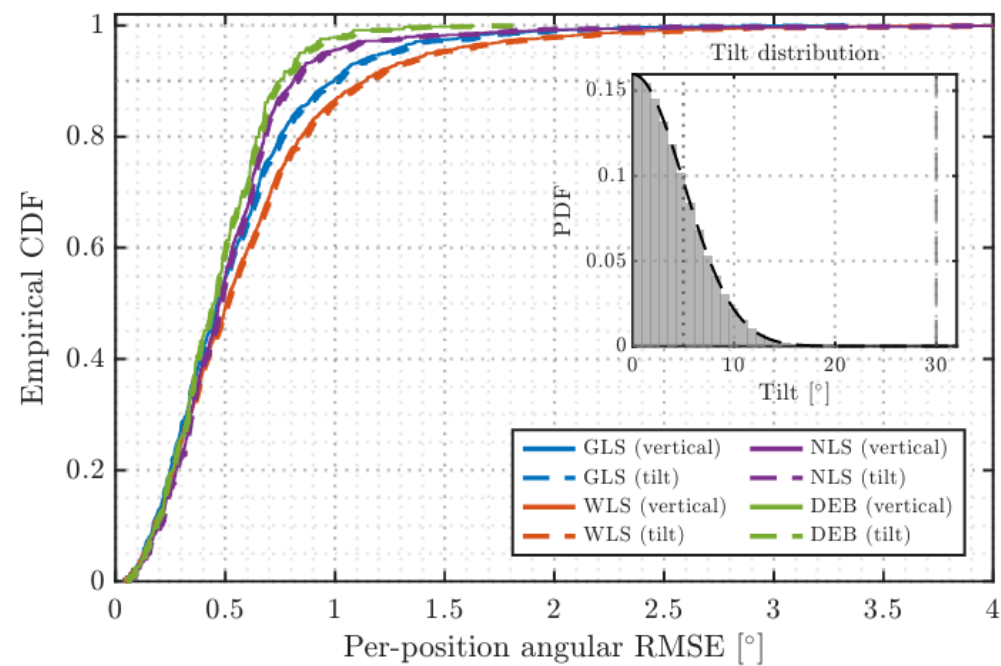


1) PhD topic:

Cramer-Rao Lower Bound

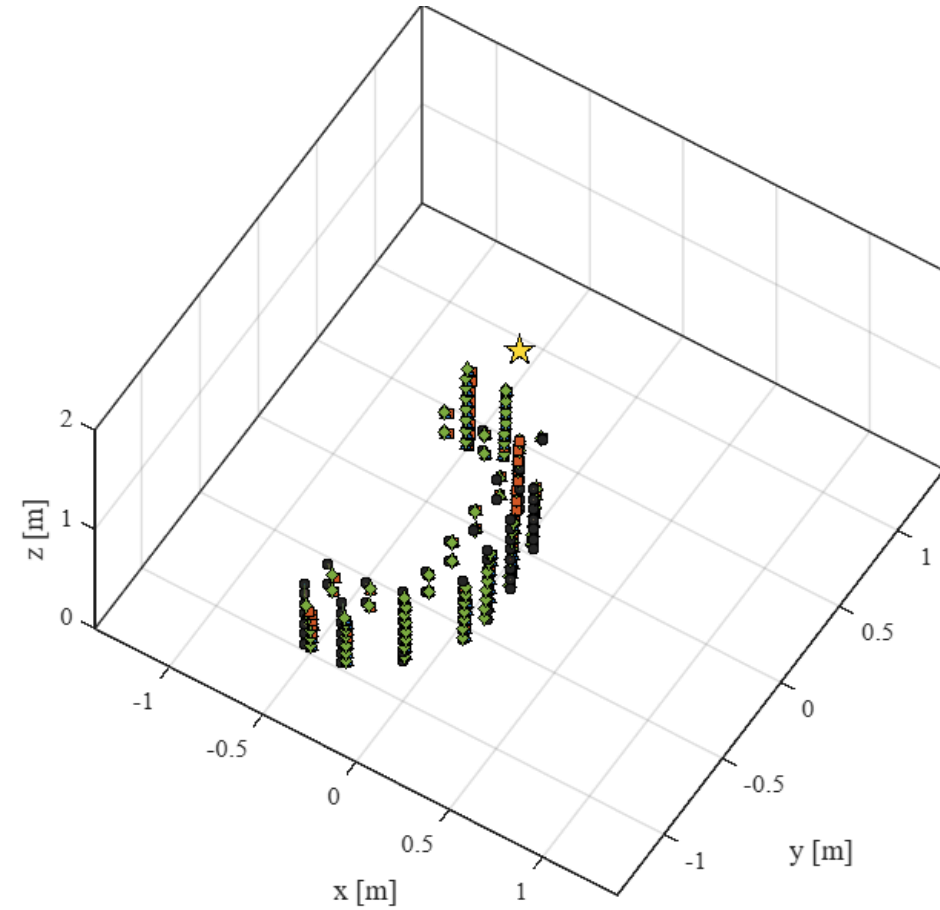
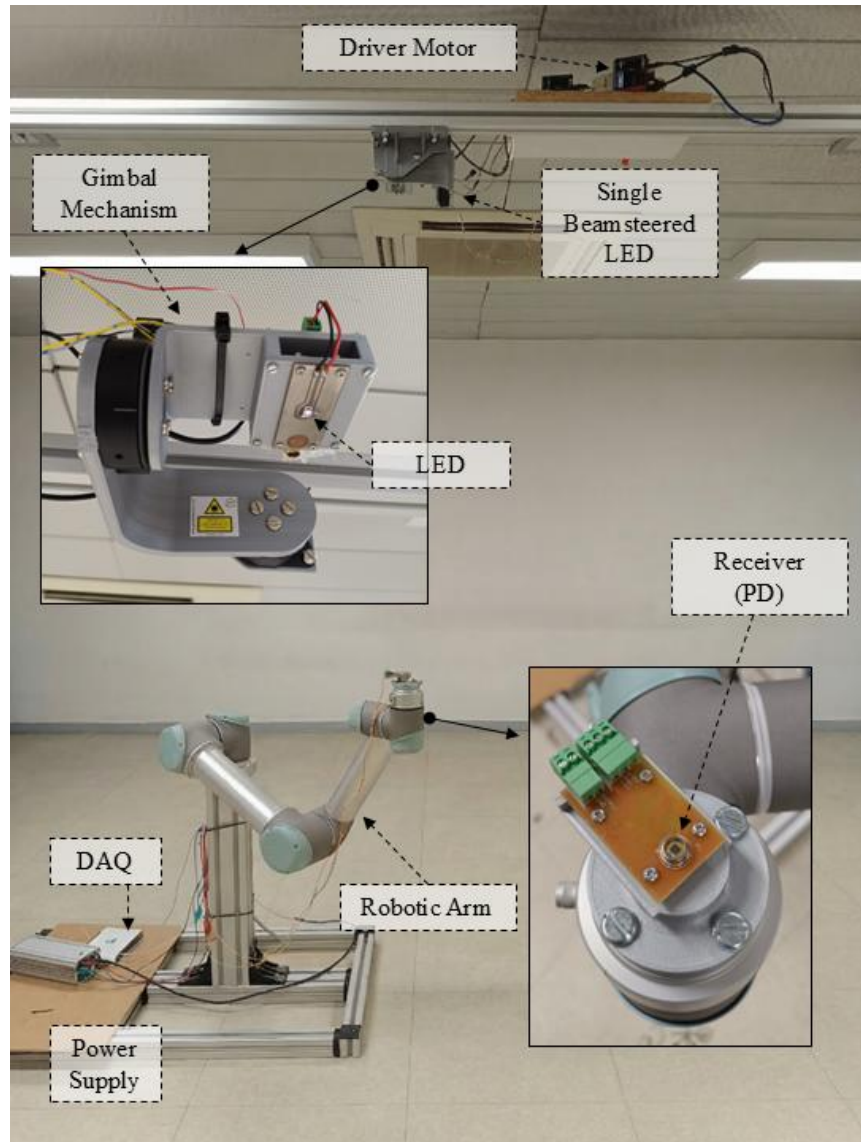


Design

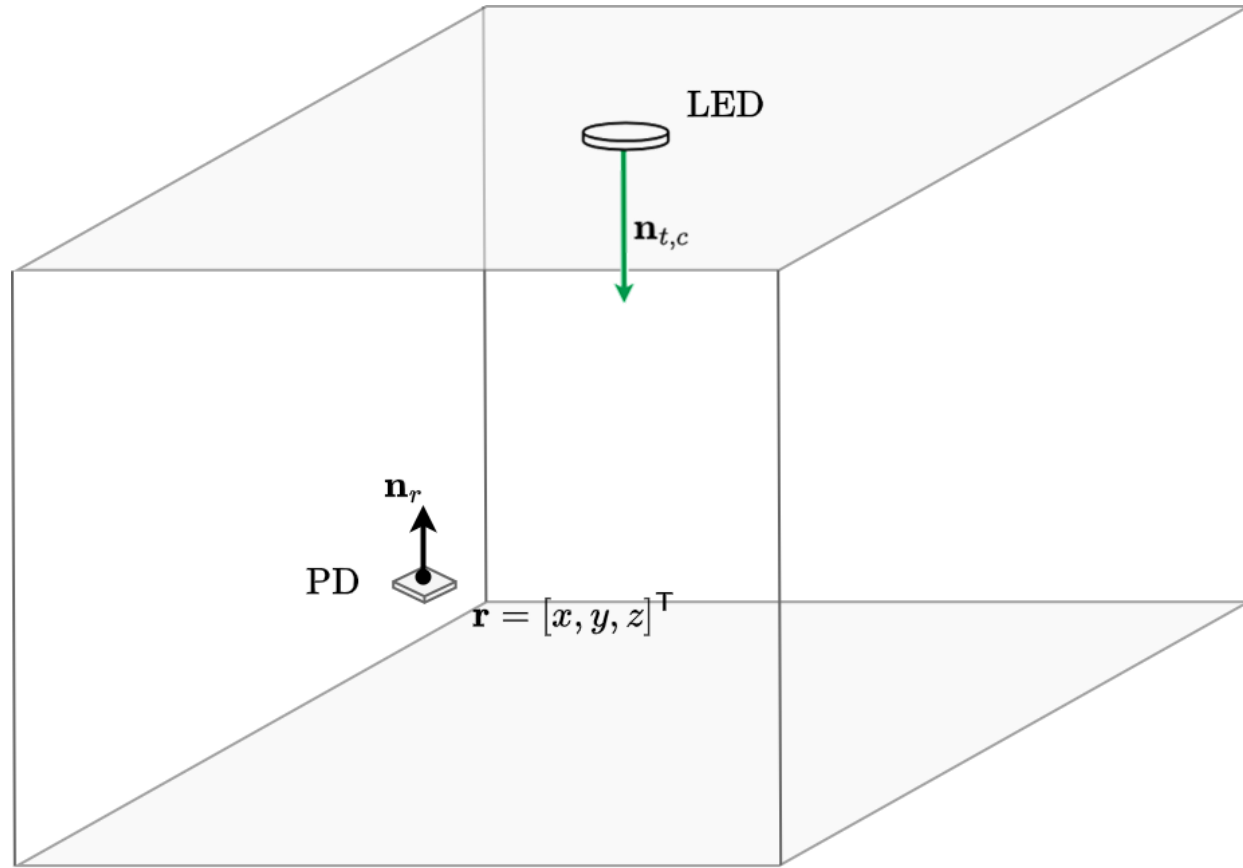


Estimators

1) PhD topic:

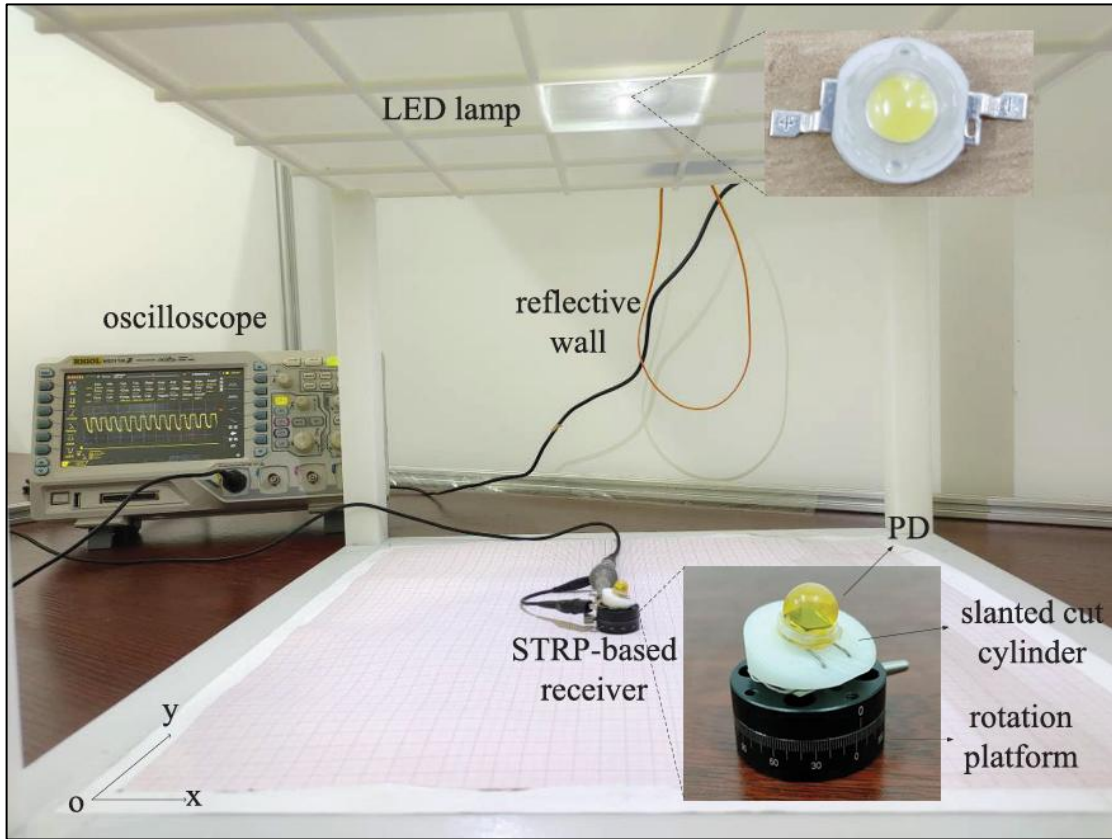


2) Challenge

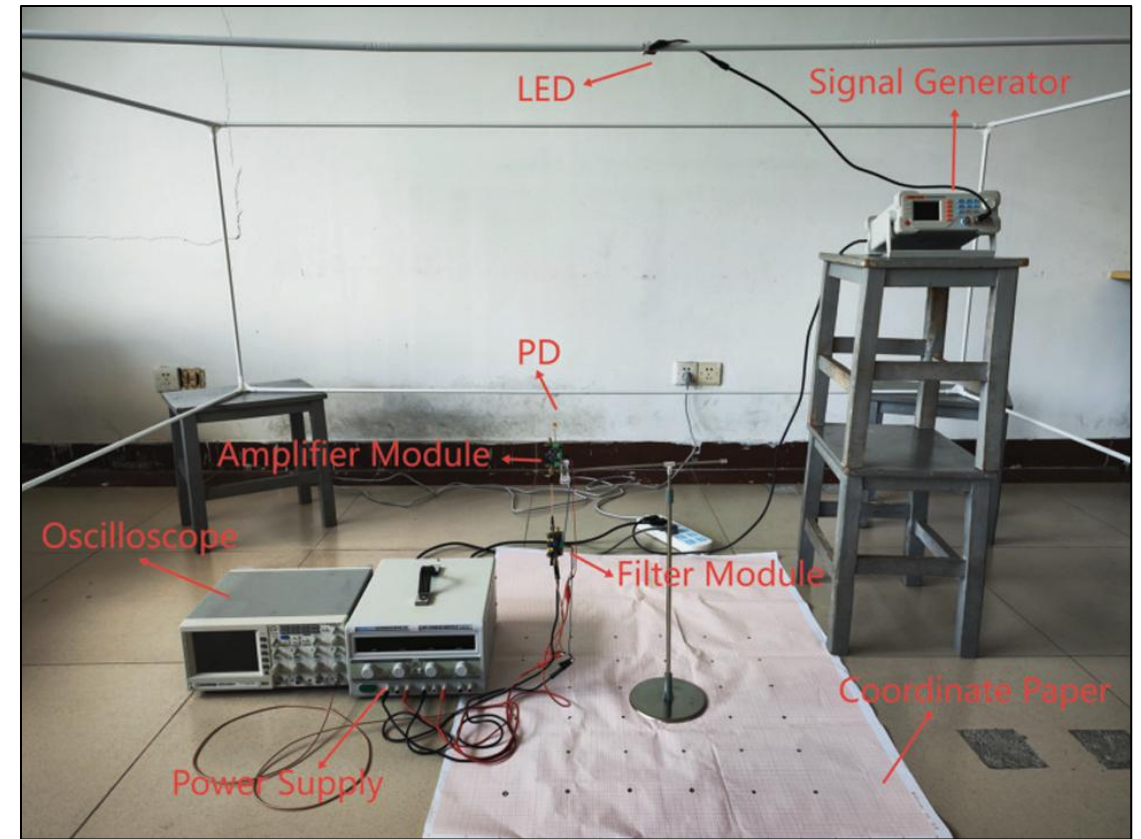


2) References

Rotatables



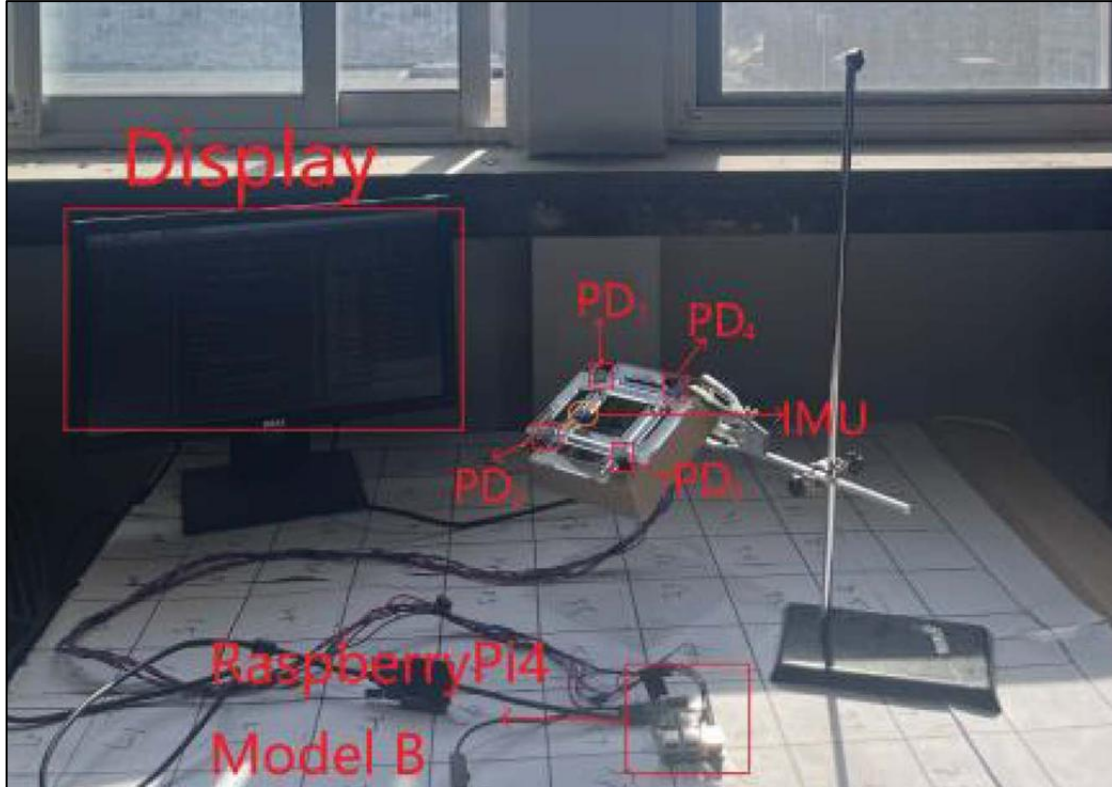
Z. Wang, 2024, “Design and Performance Analysis for Indoor Visible Light Positioning With Single LED and Single-Tilted-Rotatable PD”



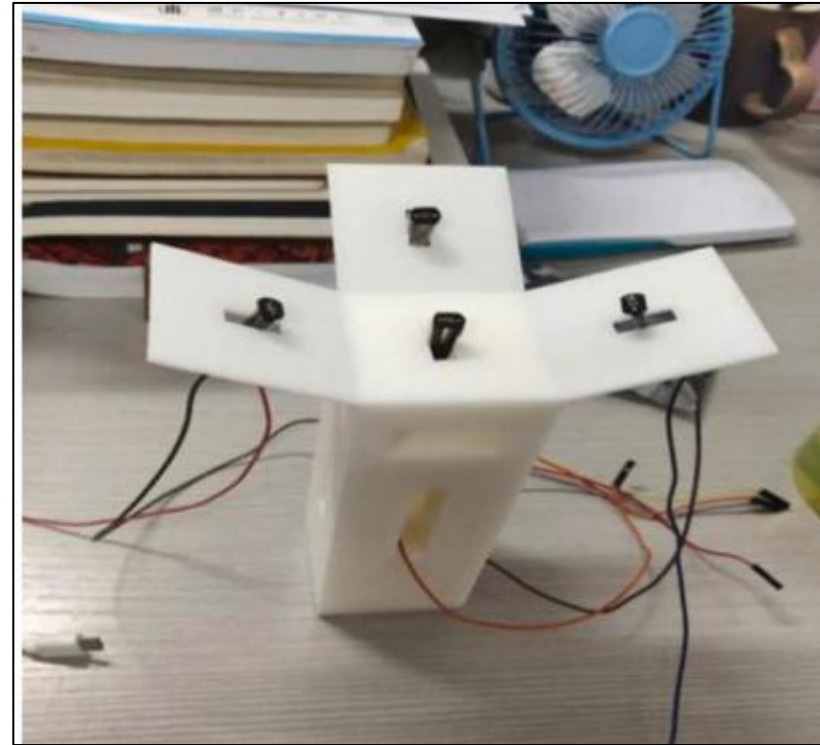
H. Shi, 2025, “Visible Light Positioning Using a Single LED and a Single Rotatable Photodetector”

2) References

Statics



S. Ma, 2024, “Centimeter-Level 3-D Mobile Online Visible Light Positioning System With Single LED Lamp”



Qin L., 2021, “High precision indoor positioning algorithm of single LED lamp based on A-Bayes”

3) Virtual testbed